

Power Electronics Laboratory

Session 1: DC-DC converter 1

Team 3 :

Po Ting Yeh K-9056

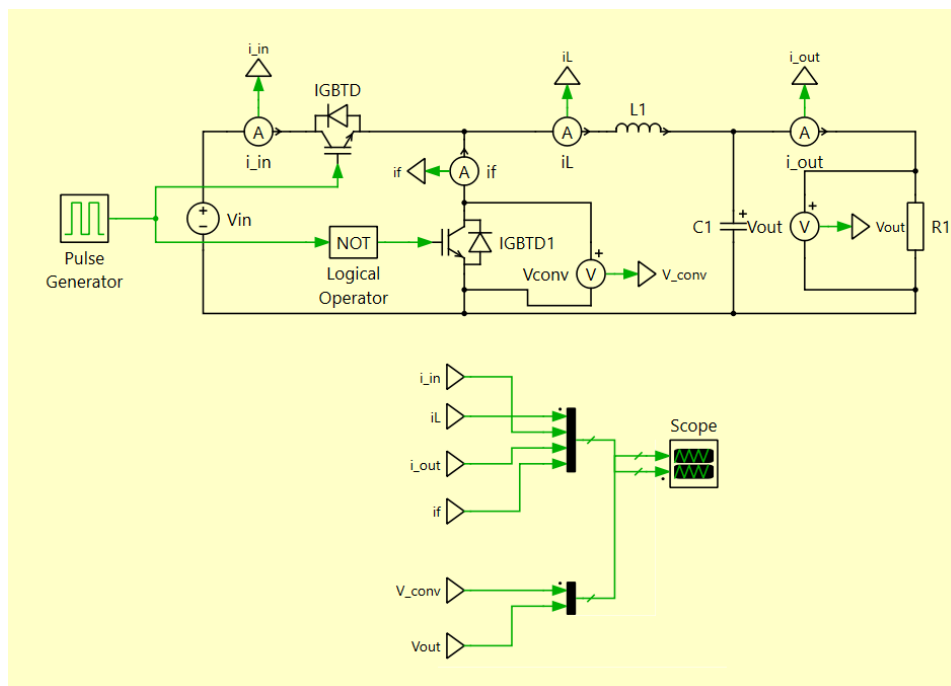
Beyazit Bilal Ozkaya 341967

Braulio Cardenas 342021

Introduction

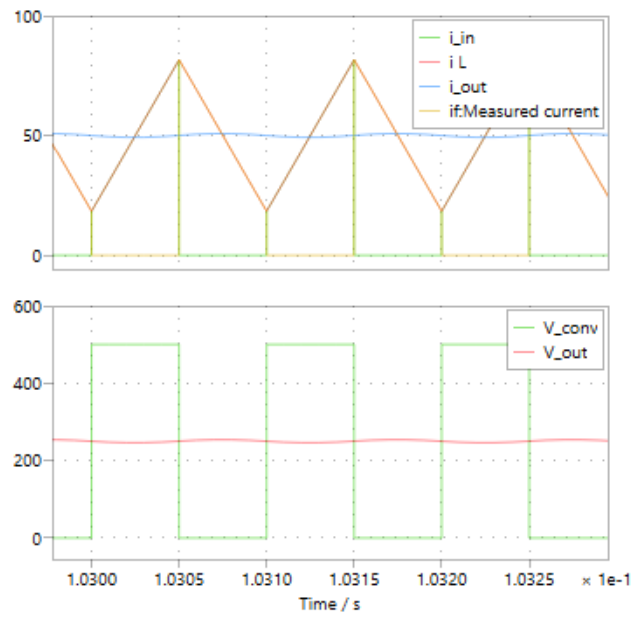
This laboratory experiment investigates the steady-state operation and switching characteristics of a synchronous DC-DC buck converter utilizing the PLECS simulation environment. The study explores the implementation of Pulse Width Modulation (PWM) to regulate the output voltage through varying duty cycles. Furthermore, the experiment assesses how modifying critical system parameters—specifically inductance, capacitance, switching frequency, and load resistance—impacts the converter's current and voltage ripples. Finally, a non-ideal converter model is examined by introducing parasitic resistances to observe conduction losses, highlighting the need for manual duty cycle compensation to maintain a regulated output voltage during load and line disturbances.

Part1

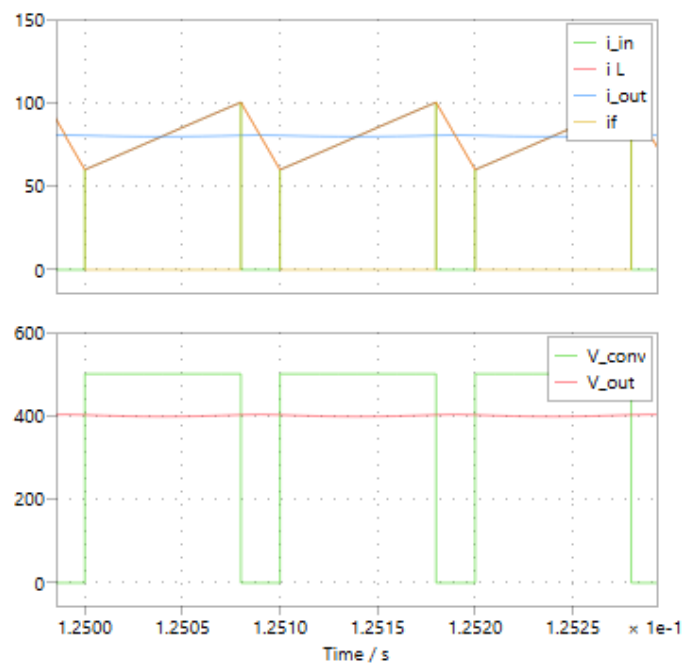


Part2

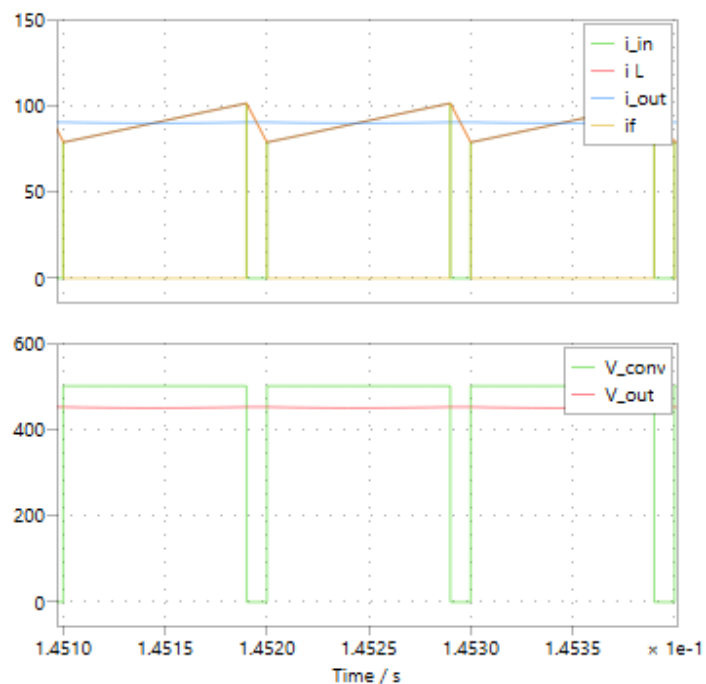
1. Duty cycle: 0.5 -> output 250V (250 V theoretical)



2. Duty cycle: 0.8 -> output 400.33V (400 V theoretical)



3. Duty cycle: 0.9 -> output 449.789V (450 V theoretical)

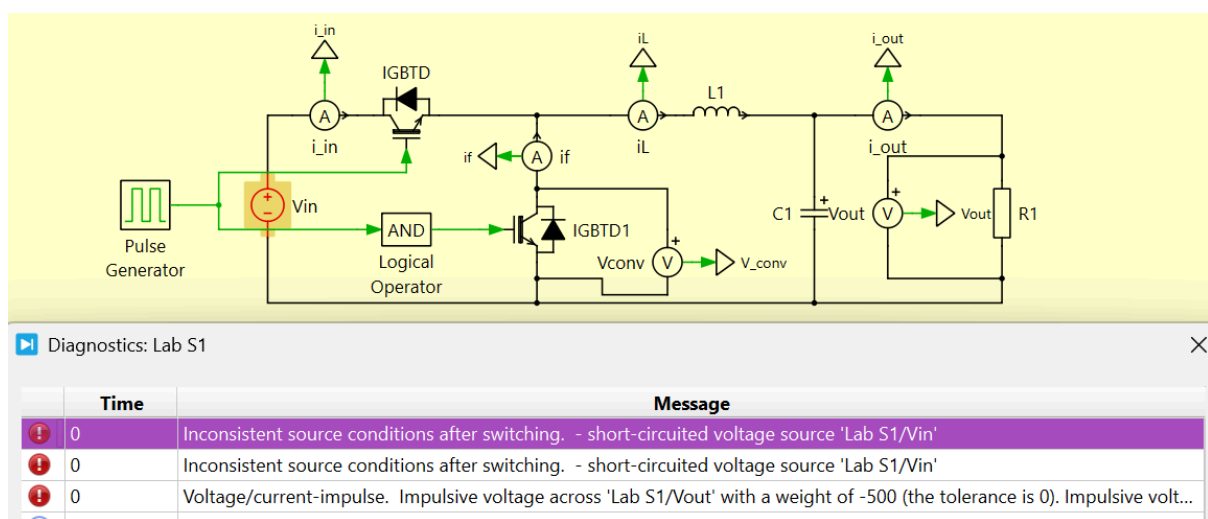


Q3: Does the output voltage match the expected value?

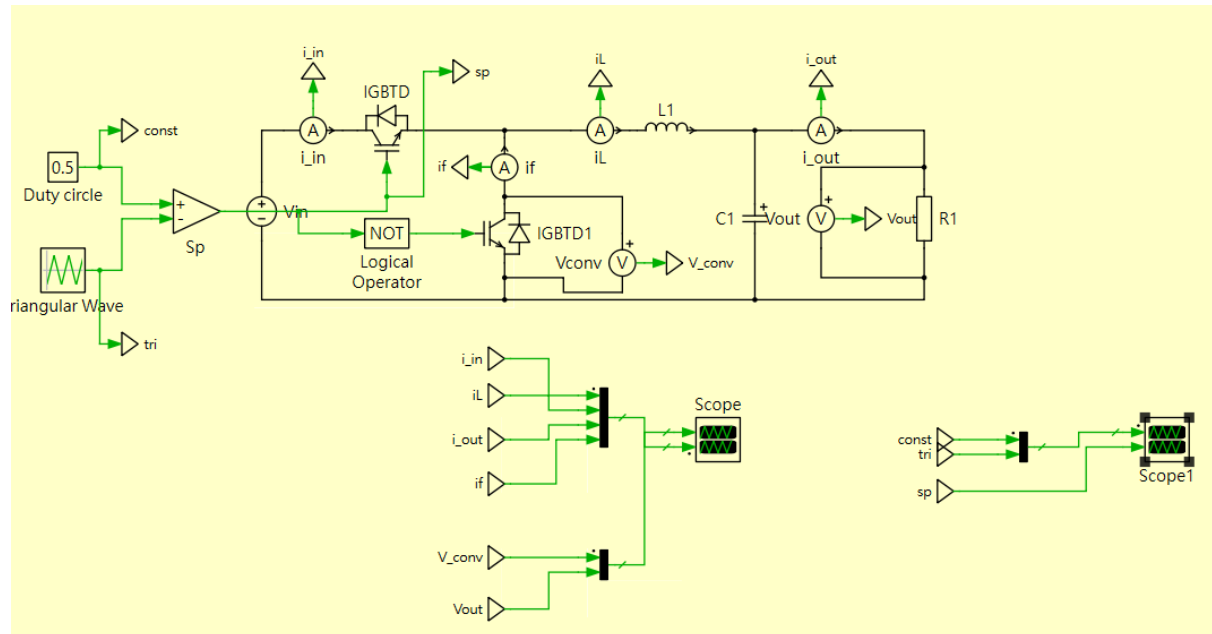
A: Yes, The output voltage simulated by PLECS almost matches the expected value.

Q4: What happens if sp and sn are activated at the same time?

A: If both switches are on at the same time, it causes a "shoot-through". This creates a short circuit between the input source and ground. In a real circuit, this would cause massive current and destroy the IGBTs due to overheating.

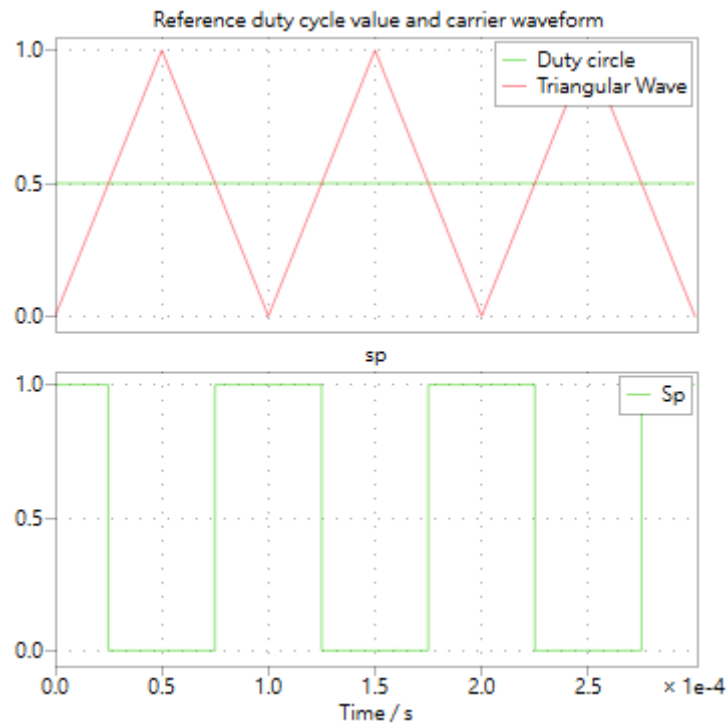


Part3

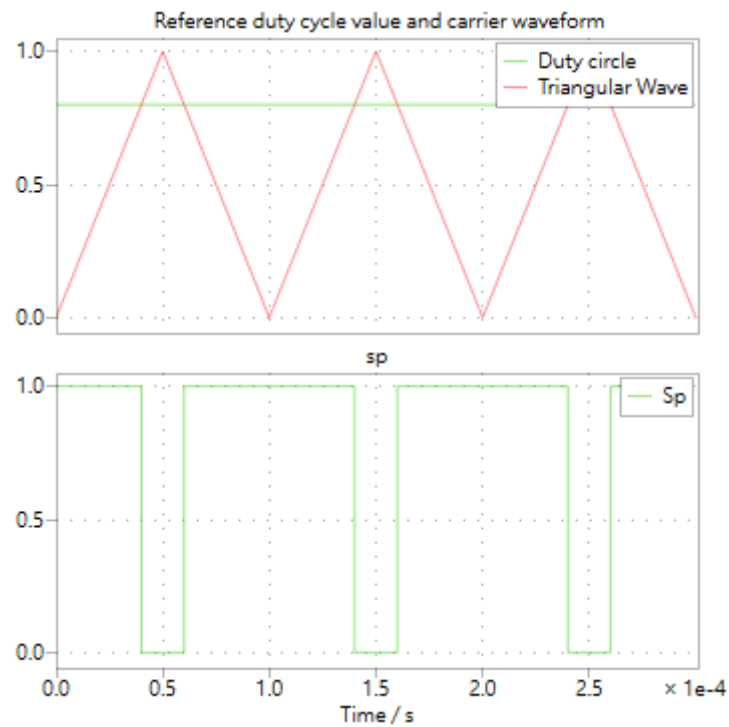


Q2. Verify the operation of the PWM generator with the duty cycle values used in Part 2.

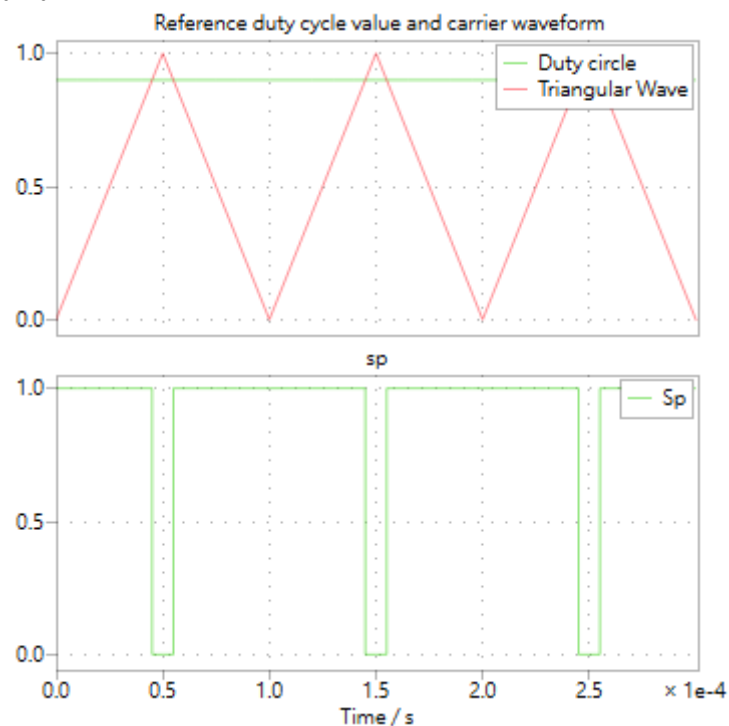
1. Duty cycle: 0.5



2. Duty cycle: 0.8



3. Duty cycle: 0.9



Q3: Calculate the average value of sp . Is it equal to the reference duty cycle? What happens with Vconv?

1. Average value of sp

The average value of sp will be 0.5, 0.8, and 0.9, depending on the value you set for the constant block.

2. Is it equal to the reference duty cycle?

Yes, the average value of the output pulse (sp) is exactly equal to the reference duty cycle (d^*).

3. What happens with Vconv?

The voltage Vconv has the exact same square waveform shape as the sp signal, but its amplitude (height) changes:

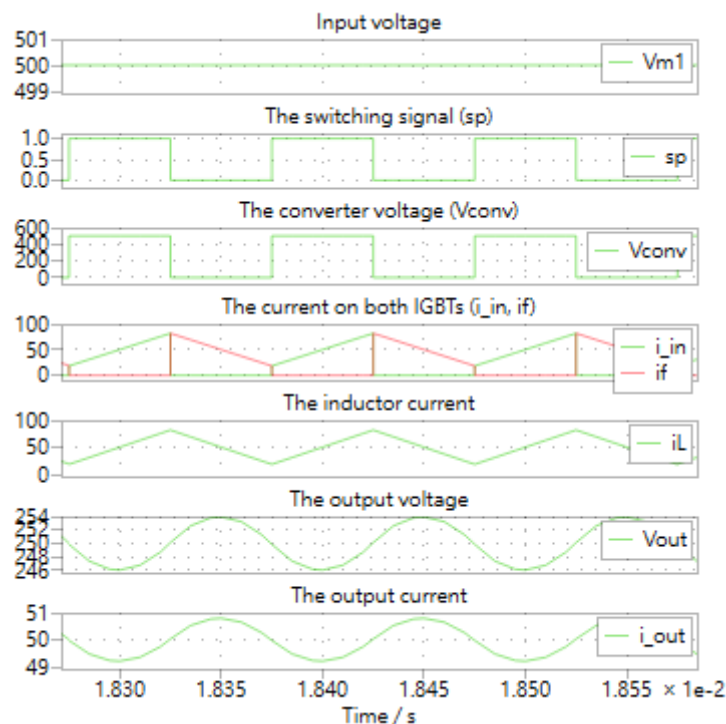
- When $sp = 1$ (upper switch is ON), Vconv equals the input voltage V_{in} .
- When $sp = 0$ (lower switch is ON), Vconv equals 0.
- Because of this, the average value of Vconv is simply the input voltage multiplied by the duty cycle.

Q4: Check if the output voltage matches the values obtained in Part 2.

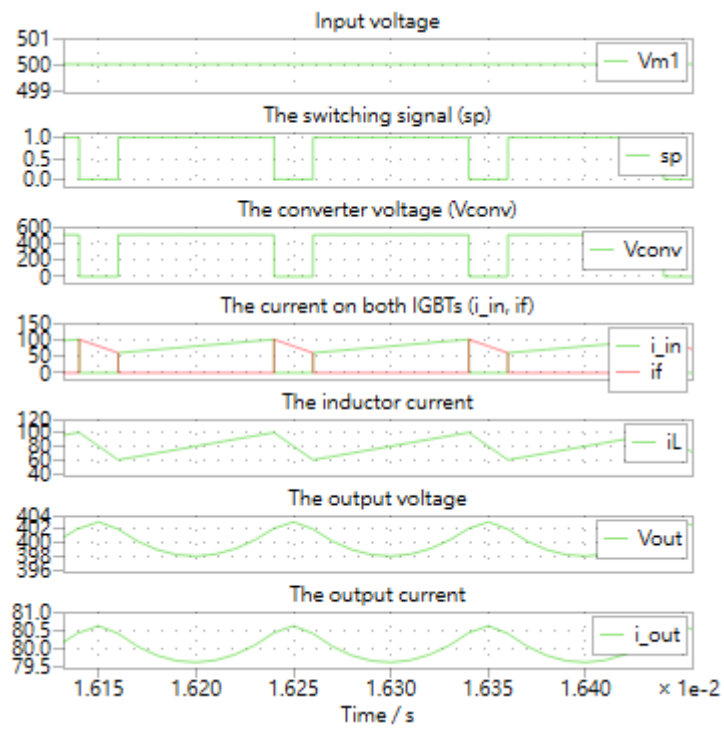
Yes, after checking the simulation result, output voltages are also 250, 400 and 450 separately.

Q5: For each duty cycle value, simulate until reaching the steady-state. Plot 3 cycles of the following variables:

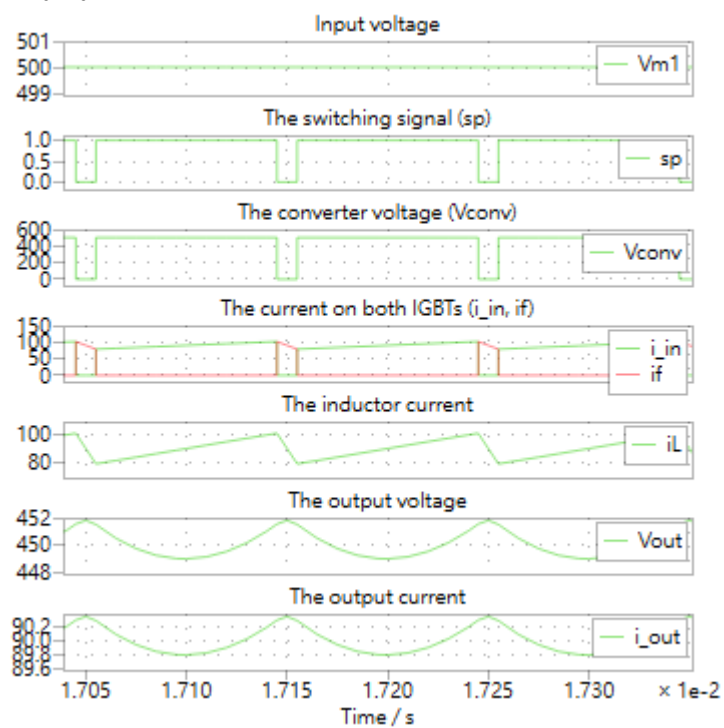
1. Duty cycle: 0.5



2. Duty cycle: 0.8

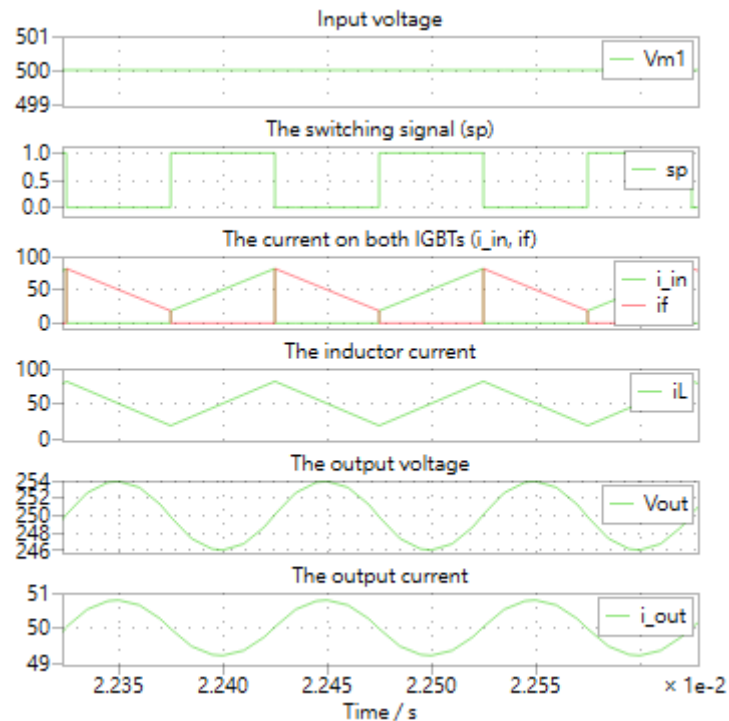


3. Duty cycle: 0.9



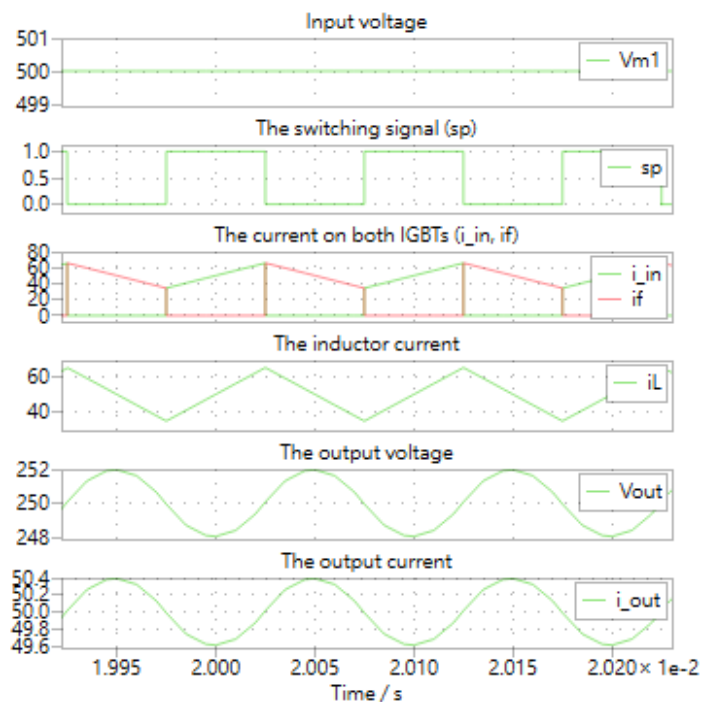
Part4

Simulation of the power converter with the parameters of Table 1:

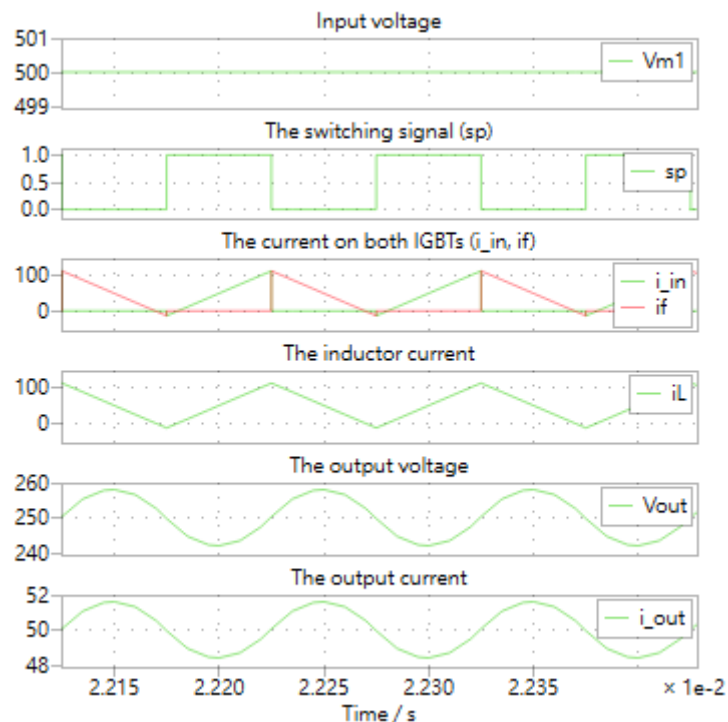


For each task, the specified parameter is **multiplied by 2 (increased)** and **divided by 2 (decreased)** from its original value in Table 1 to clearly observe the changes in the system's steady-state behavior.

1. Increase and decrease the value of the inductance, L

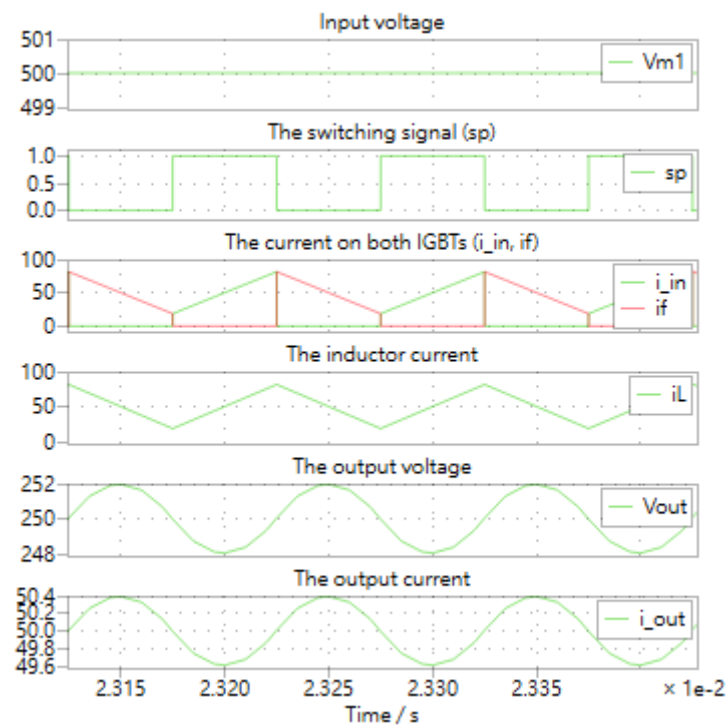


Increased (2L): The inductor current ripple decreases, resulting in a flatter current waveform.

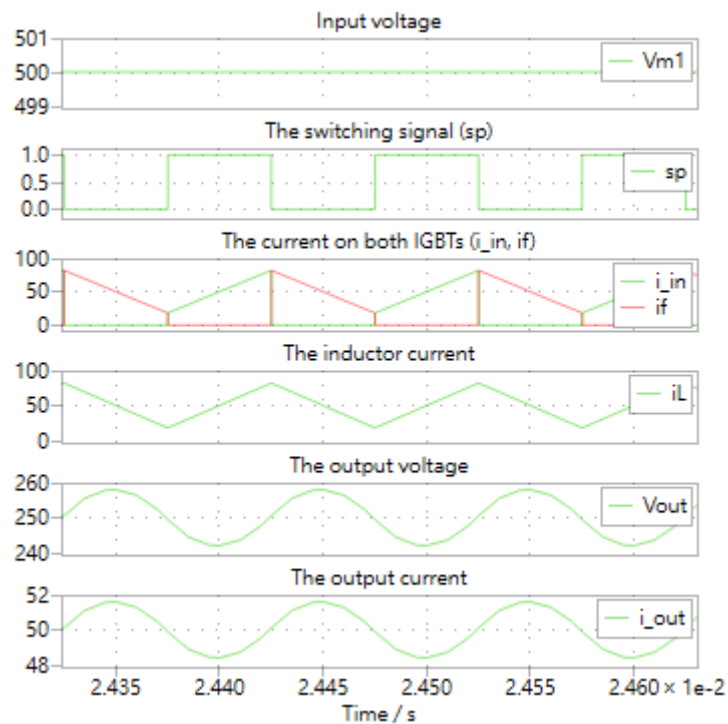


Decreased (0.5L): The inductor current ripple increases significantly.

2. Increase and decrease the value of the capacitance, C

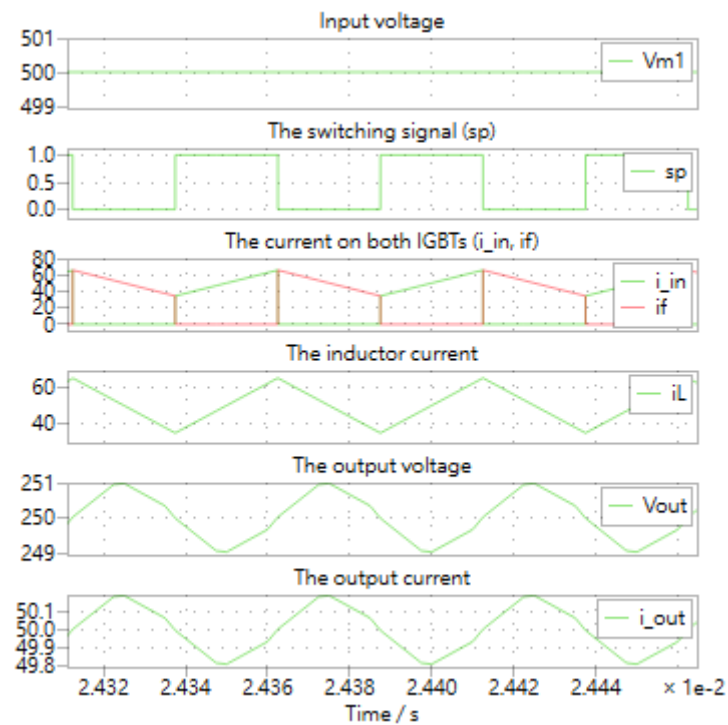


Increased (2C): The output voltage ripple decreases, making the output voltage much smoother.

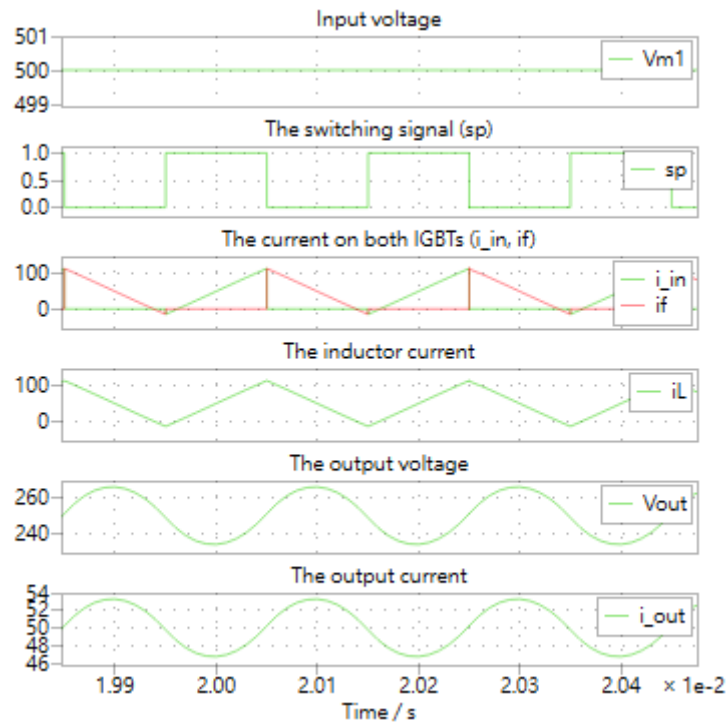


Decreased (0.5C): The output voltage ripple increases. (The inductor current ripple remains mostly unaffected).

3. Increase and decrease the value of the switching frequency

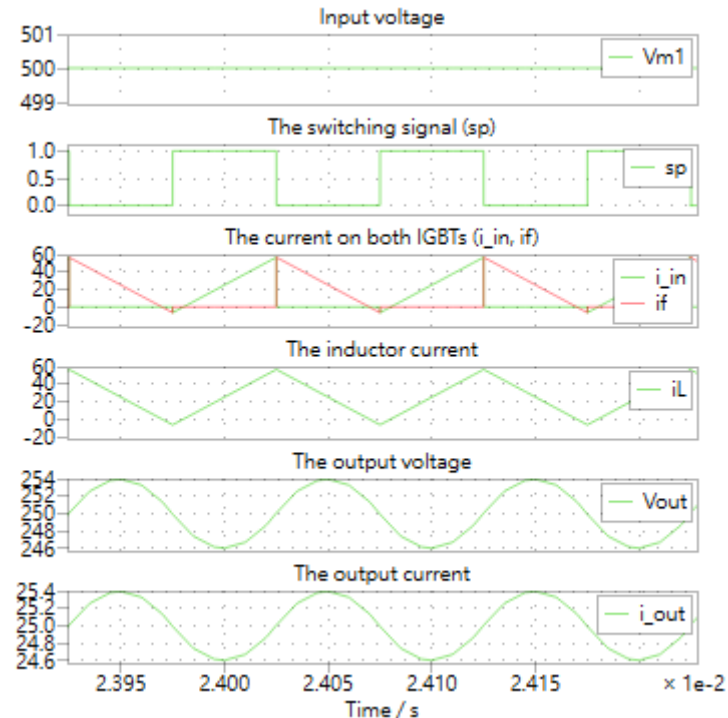


Increased (2fs): Both the inductor current ripple and the output voltage ripple decrease because of the shorter charging and discharging periods.

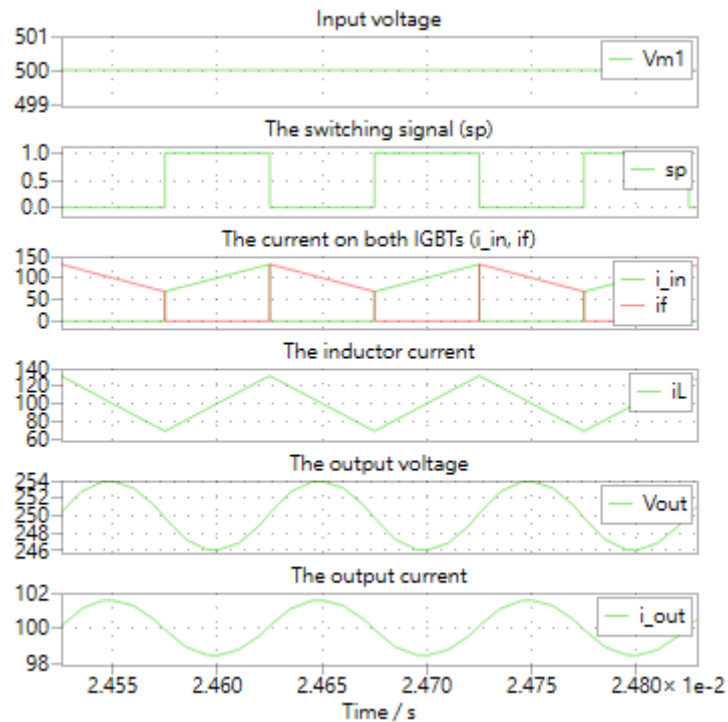


Decreased (0.5fs): Both the current and voltage ripples become noticeably larger.

4. Increase and decrease the value of the load resistance

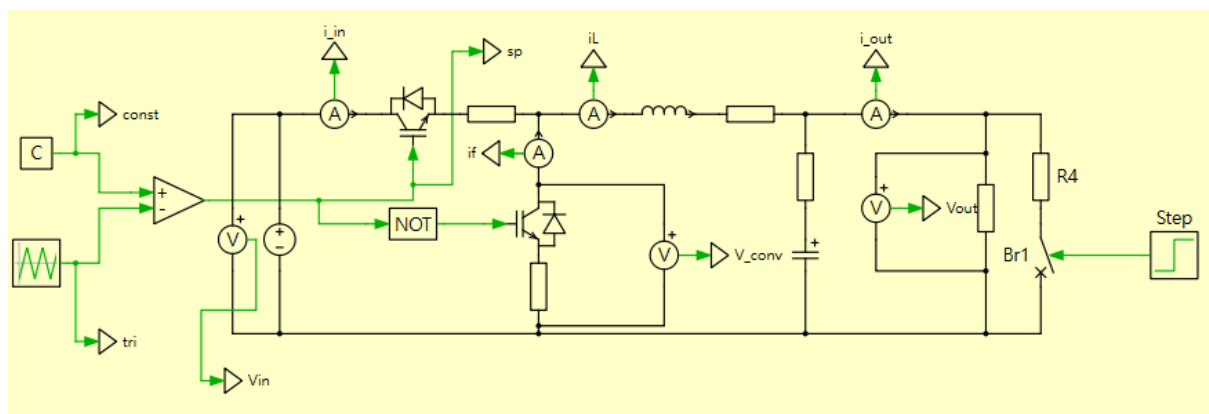


Increased (2R): The average output current and average inductor current decrease. The magnitude of the current ripple remains unchanged.



Decreased (0.5R): The average output current and average inductor current increase.

Part5



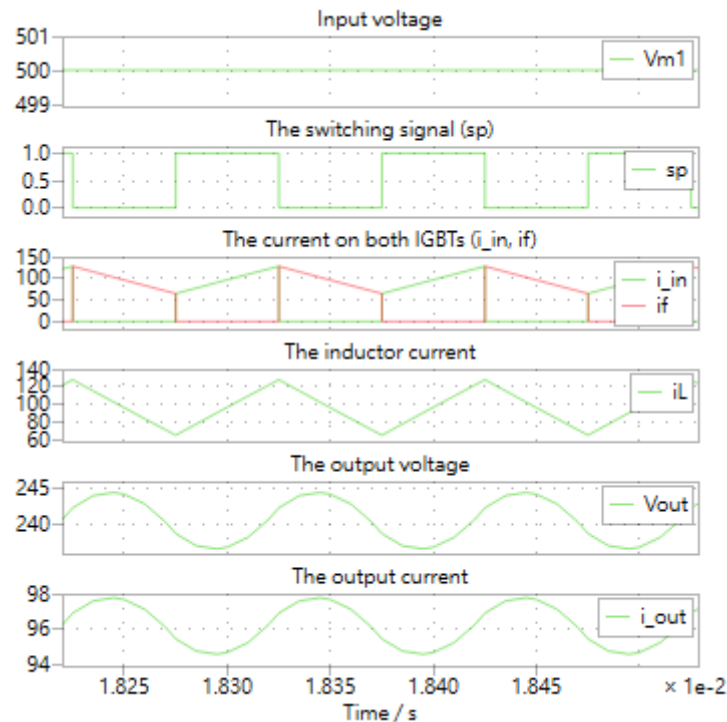
Schematic with Rloss

Q3: Compare the output voltage (With and without losses).

For a duty cycle of 0.5 and an input voltage of 500V, the ideal synchronous buck converter produces an average output voltage of 250V. However, after adding the loss resistors, the average output voltage drops noticeably below 250V which is around 240V

This voltage drop occurs because the inductor current flows through the parasitic resistances of the switches and the inductor, causing conduction losses. As a result, a

portion of the power is dissipated as heat, reducing the actual voltage delivered to the load. Furthermore, adding the capacitor's Equivalent Series Resistance (ESR) increases the peak-to-peak output voltage ripple.



Simulation with R_{loss}

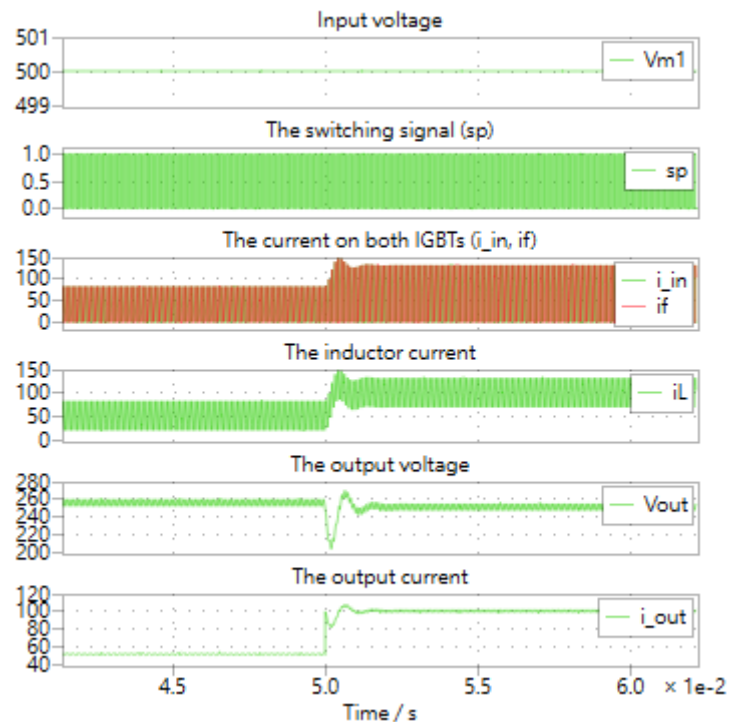
Q4: Adjust manually the duty cycle to obtain the same output voltage as in the ideal case. For the ideal and nonideal cases.

A: In the ideal case with $D = 0.5$, the average output voltage was exactly 250V. In the non-ideal case, the introduction of parasitic resistances caused a noticeable voltage drop across the components. To compensate for these conduction losses and restore the output voltage to the target of 250V, the reference duty cycle had to be manually increased. Through manual tuning, a duty cycle of **approximately 0.522** was required to achieve the 250V output. This demonstrates that in practical power converters, a higher duty cycle is necessary to overcome internal power losses and supply the desired voltage to the load.

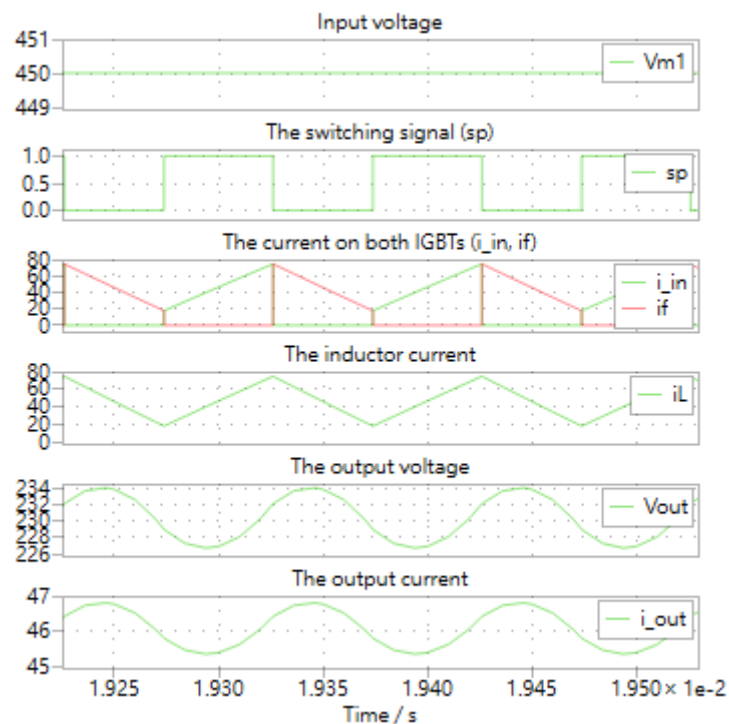
Q5: With the modified duty cycle obtained in the previous task: Simulate the system, and once the steady state is achieved, increase the output current by connecting the parallel resistance shown in Fig. 4. Use the electrical switch and configure the switching time accordingly. Show the transient response.

A: A parallel resistor was connected to the output using an electrical switch configured to close at $t = 0.05$, after the system had reached steady state. When the switch closes, the total load resistance decreases, causing a sudden step increase in the output current. Because the converter operates in an open-loop configuration with a fixed duty cycle, the increased load current flowing through the parasitic components R_{loss} results in a

larger voltage drop ($V_{\text{drop}} = I \cdot R_{\text{loss}}$). Consequently, the transient response shows the output voltage sagging and eventually settling at a new, lower steady-state value. This demonstrates the limitation of open-loop control when load variations occur.



Q5: With the modified duty cycle of task 4, reduce the input voltage to 450V. Check the new output voltage. Again, modify the duty cycle to obtain the desired output voltage.



The input voltage was reduced from 500V to 450V while keeping the previously adjusted duty cycle. As expected, this input voltage disturbance caused a proportional drop

in the output voltage, **which fell to approximately 230.81V**. To compensate for both the lower input voltage and the internal conduction losses R_{loss} , the duty cycle had to be manually increased again to regulate the output voltage back to the target of 250V. Through iterative adjustment, a new duty cycle of **0.568** was required to achieve this.

Q6-1: Why the output voltage is affected when increasing the load current?

A: The output voltage decreases when the load current increases due to the presence of parasitic resistances R_{loss} in the components, such as the inductor's copper resistance and the switches' conduction resistance. According to Ohm's law, a higher current flowing through these internal resistances creates a larger voltage drop ($V_{drop} = I * R_{loss}$). Since this voltage is lost as heat inside the converter, the remaining voltage delivered to the output load is reduced.

Q6-2: What happens with the inductor current and semiconductor currents?

A: When the load current increases, the average inductor current must increase proportionally to supply the higher demand of the load. Consequently, the average currents flowing through both semiconductors (the upper IGBT during the ON-state and the lower IGBT/diode during the OFF-state) also increase. Graphically, the entire current waveforms (i_L , i_{in} , I_f) shift vertically to a higher average level, while their peak-to-peak ripple magnitude remains largely unchanged.

Conclusion

The simulation results successfully verify that the output voltage of an ideal synchronous buck converter is directly proportional to the reference duty cycle of the PWM signal. Parameter variation analysis demonstrates that increasing the inductance or capacitance effectively reduces the inductor current ripple and output voltage ripple, respectively, while a higher switching frequency simultaneously mitigates both ripples. The introduction of parasitic resistances reveals that internal conduction losses cause a measurable voltage drop, reducing the actual voltage delivered to the load as heat dissipation. Consequently, when subjected to load increases or line voltage reductions, the open-loop converter suffers from steady-state errors, emphasizing the necessity of closed-loop control systems to automatically adjust the duty cycle and maintain a fully regulated output.